

Nimble RX

Proposal | Nimble @ WashU Robotics, Fall 2026

Problem

Prescription handling is largely manual: pharmacy staff count, label, and verify orders by hand, and patients/caregivers sort weekly medications into organizers, a repetitive, human error-prone task. Nimble RX explores a natural-language-initiated prescription handling pipeline built on Vision-Language-Action (VLA) models to fix the human risk of misplacement.

Solution

Pipeline

```
"Put the red prescription in Container A."  
-> Whisper STT -> schedule parser (local Qwen LLM)  
-> [(rx_i, container=i), ...]  
-> Kokoro TTS read-back confirmation  
-> for each (rx, container):  
    -> VLA Inf. #1: pick -> collision check: pass -> continue | fail -> withhold  
    -> VLA Inf. #2: place -> collision check: pass -> continue | fail -> withhold  
    -> YOLO CV model: correctness check -> pass -> done | fail -> withhold
```

Implementation

- **Data collection.** LeRobotDataset episodes (instruction, camera frame, 6-DOF grasp/place chunk) per item, initial set comprises 10 episodes/type, scaling toward ≥ 50 /type. Isaac Lab may accelerate collection.
- **Safety.** A YOLO CV model checks correct-container placement, serving as a post-placement evaluator and safety outcome predictor.
- **Training.** SFT on NORA 1.5 [1] via its official fine-tuning script, trained/evaluated per-item to keep the success signal dense. *Stretch:* DPO post-training, pairing successful vs. failed rollouts of the same instruction using CV verdicts as preference labels.

Equipment

Arm	WidowX AI Base (6-DOF)	Organizer	Containers for related prescription items
Gripper	WidowX squeeze gripper	Items	10 3D-printed prescription replicas
Cameras	Intel RealSense D435i	Compute	RoboServer GPU inference
Base	Linear actuator (workspace)	Env.	Fixed, well-lit, curtained safety boundary

Key Deliverables

- **LeRobotDataset** of synchronized frames + user input + grasp/place action chunks, a **fine-tuned VLA** (NORA 1.5) for prescription pick-and-place, a **linear actuator** extending the arm's workspace, a **YOLO CV model** for safety and potentially a DPO reward, finally a **public release** of dataset, model, metrics, videos, and report on HuggingFace / GitHub.

Timeline

Wks 1–3	Equipment sourcing (3D printing, ordering), environment setup, RoboServer setup
Wks 4–7	Per-item data collection, dataset build, preprocessing pipeline.
Wks 8–10	Fine-tuning, CV model, safety guardrails.
Wks 11–12	Evals, documentation, public release.

Success Metrics

- **Per-item success rate:** $\geq 80\%$, measured over 5 attempts per item type.
- **Safety effectiveness:** $\geq 80\%$ of unsafe predicted actions correctly withheld.

References

[1] Declare Lab. *NORA 1.5 Model Card*. Hugging Face, 2025.